# Spend Pixels Where It Hurts: Hazard-Aware Rendering for Optical Long-Context Compression

Anonymous CVPR submission

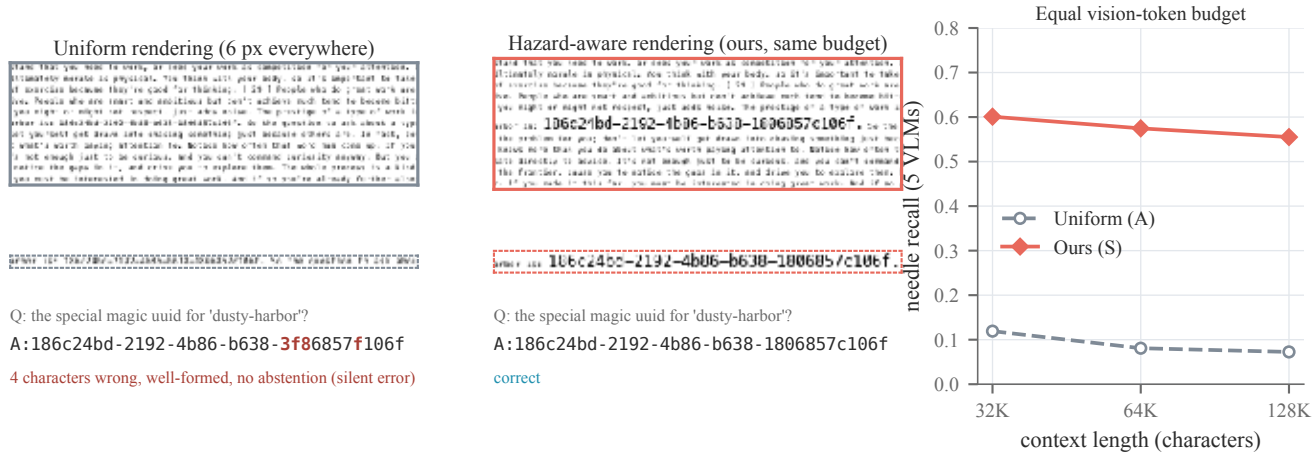



Figure 1. **Uniform rendering silently corrupts exactly the strings that must be copied.** A 32K-character essay hides one UUID. *Left:* rendered uniformly at 6 px, Claude Sonnet 5.5 returns a well-formed UUID with four wrong characters and no sign of doubt. *Middle:* our renderer enlarges the UUID within the same vision-token budget (here the prose keeps its 6 px size), and the answer is exact. *Right:* needle recall averaged over five VLMs and three RULER tasks; uniform rendering degrades with context length while hazard-aware rendering does not.

## Abstract

*Rendering long text into images lets vision-language models (VLMs) read it with several times fewer tokens than the text itself. This optical compression spends resolution uniformly, yet the strings that must be copied exactly, such as identifiers and hashes, are misread into well-formed but wrong answers far more often than prose, and they fail more often as contexts grow. We ask where the pixels should go. Because the renderer, unlike the model, sees the text, it can flag high-entropy strings and draw them larger while drawing prose smaller, within the vision-token budget of uniform rendering. This allocation is training-free, query-independent, and needs only black-box access to the VLM. Across five commercial and open-weight VLMs accessed through APIs, it raises RULER needle recall over uniform rendering by 0.55 and 0.43 at the two densest budgets (95% CIs [0.48, 0.62] and [0.27, 0.56]) and by 0.48 at 32K–128K characters, while using up to 27% fewer billed tokens. Enlarging random words, or boxing the needles without enlarging them, does not help, so the gain comes from legibility of the targeted strings. The gain shrinks as the budget relaxes and is no larger than a font search at 10 px, and on LongBench it doubles the recall of naturally occurring identifiers that the detector flags (0.23 to 0.48) without changing QA accuracy relative to uniform rendering; strings the detector misses gain nothing. For flagged targets, verbatim anchors from prior work are as accurate on the allocated image as on a uniform one, and cheaper wherever allocation shrinks the image. All image arms, however, remain well below plain text on ordinary QA, so allocation makes optical compression safer for identifiers rather than a substitute for text.*

## 1. Introduction

Long contexts are expensive to feed to large models. A growing line of work reduces this cost by *rendering* text into images and letting a vision-language model (VLM) read the pixels [8, 34, 41]. A page of small glyphs costs a fixed number of vision tokens regardless of how many charac-

ters it holds, so dense rendering can compress a document several-fold. In our measurements, a 128K-character essay costs 27.1K billed text tokens but only 4.7K billed tokens as a uniformly rendered image (mean of the four readers that report billed usage for both). The compression needs no training, works with any VLM that accepts images, and is controlled entirely by the party that renders the text.

Existing systems decide *how densely* to render, but rarely *where* the resolution goes. Optical compressors choose one font, size, and layout for the whole document, sometimes searched for a good accuracy–compression tradeoff [8]. Methods that do vary resolution within the input are trained [44, 46, 53] or read the model's attention for each query [48], and visual-token reduction prunes or merges tokens after encoding [6, 47]. None of these is available to a party that only renders the text and calls a black-box API.

The cost of uniform resolution is not uniform. Identifiers, hashes, file paths, and long numbers carry no redundancy, so a single misread glyph turns them into a different, equally plausible string, while the surrounding prose survives the same blur. Figure 1 shows the typical outcome: a VLM returns a well-formed UUID with four wrong characters and no hedge. Prior work on failure-aware optical compression [1] showed that such silent corruption concentrates in high-entropy spans, and responded at the level of whole blocks: it flags these spans from the text, copies them verbatim next to the image as *anchors*, and routes risky blocks back to text. These remedies leave the image itself unchanged. We confirm the problem at long-context scale: at an equal budget, uniform rendering recovers only 8–22% of RULER needles (mean of five readers, Tab. 1), and recall falls from 0.12 at 32K characters to 0.07 at 128K.

Our observation is that the renderer already holds the signal needed to fix the image. Before any pixel is drawn, the text says where the hazardous strings are, and the renderer decides how many pixels each of them gets. We therefore propose *hazard-aware rendering* (Fig. 2): flagged strings are drawn larger and prose smaller, under the constraint that the image costs no more vision tokens than uniform rendering. The allocation (arm S in our experiments) is training-free and query-independent, so the same rendering can serve every question about a document. It composes with the anchors and text routing of prior work [1], which we treat as baselines and building blocks (arms K and X).

Our contributions are:

- **Budget-constrained glyph allocation** for rendered context, a rule that redistributes resolution inside the image using only the text and black-box access to the reader (Sec. 3).
- **Controlled evidence for where it helps.** Across five readers, allocation raises needle recall over uniform rendering by 0.43–0.55 at 16K characters and by 0.48 at 32K–128K, and beats font/size search, render-then-downsample, enlargement of random words, and boxing the needles without enlarging them (Tab. 2). The gain shrinks at milder budgets and matches a font search at 10 px.
- **Natural strings and side effects.** On identifiers in LongBench documents that the detector flags, allocation doubles recall (0.23 to 0.48); on digit-bearing strings sampled independently of the detector, which it mostly does not flag, it neither helps nor hurts. It leaves LongBench QA unchanged relative to uniform rendering, while every image arm, ours included, trails plain text.
- **Composition with anchors.** Anchors are as accurate on the allocated image as on a uniform one (pooled X−K between −0.01 and +0.05 in every setting with flagged targets), and cost less wherever allocation shrinks the image. We also show that on standard needle tasks anchors contain the answer, so recall there measures copying rather than reading, and that anchors mislead readers when the target is not flagged.

## 2. Related Work

**Optical context compression.** Reading text from pixels has a long history in representation learning, from pixel-based language models [12, 34] to unified image–text encoders [37] and screenshot pretraining [21]. Recent systems use rendering explicitly to compress long contexts: DeepSeek-OCR decodes text from a small number of vision tokens [41] and its successor reorders visual tokens causally [42], Glyph renders long inputs and searches for a good global rendering configuration [8], rendering a whole input as one image can roughly halve its token cost [26], and other work renders distant context or interleaves rendered text with images [39, 45]. Evaluations have started to show where this fails: compressed reading loses retrieval ability [52], accuracy collapses when linguistic priors are removed [13, 27], the text–pixel gap depends strongly on font and resolution [36], and optical compression can underperform simple alternatives such as mean pooling [20]. Several methods vary resolution within the input. FocusVTC combines a low-resolution global view with adaptively enhanced regions [53], SEER learns query-dependent selective compression [46], and LensVLM post-trains a reader to expand relevant compressed pages on demand [44]; in document understanding, DocOwl2 and TextMonkey compress the visual tokens of high-resolution pages with learned modules [17, 29]. Closest to ours, AGAR re-renders a page with the word spans that receive the most middle-to-late-layer attention enlarged [48]; it is training-free but needs the model's attention and a first pass for each query. Our allocation needs neither: it is decided from the text before any model call, works with black-box APIs, and does not depend on the question, so one rendering serves every question about a document (we do not measure provider-side

caching). Failure-aware optical compression [1] is the closest prior work. It detects high-entropy spans from the text, keeps them as exact text anchors, routes risky blocks to text, and retries on mismatch, but leaves the rendered image unchanged; we change the image and reuse its anchors and routing as baselines and building blocks.

**Training-free visual token reduction.** A complementary line of work reduces the tokens of an already encoded image: FastV drops low-attention tokens after an early layer [6], SparseVLM uses text guidance [50], VisionZip and PruMerge keep dominant or merged tokens [35, 47], ToMe merges similar tokens [5], and recent methods balance importance and diversity or target the output objective [11, 30, 49]. RTPrune adapts pruning to DeepSeek-OCR by exploiting text density [38], while FCoT-VL shows that text-rich images need dedicated compression [23]. These methods operate inside the model on visual features, where it is no longer explicit that a group of glyphs forms a hash, and most need access to the model. Our method acts before encoding, from the text; whether pruning keeps or discards the enlarged glyphs is an open question that we do not test.

**Text-side and memory compression.** Prompt compressors drop tokens judged uninformative by a small language model [18, 19, 24, 33], learned compressors map context to a few soft tokens [9, 14, 32], and KV-cache methods evict entries during inference [10, 25, 43, 51]. Selective Context, like our detector, uses an information-theoretic signal, but uses it to *remove* low-information text. Verbatim strings are high-information and must not be removed; we keep every character and only change how large it is drawn. For repeated queries over a fixed document, reusing cached prompt segments [15] is a competing way to cut cost that we do not evaluate.

**Evaluation and input resolution.** We measure verbatim fidelity with RULER needle tasks [16], whose UUID and number values are exactly the strings at risk, and measure collateral damage on LongBench QA [4]; text-rich and long-document benchmarks [28, 31] document the same weakness in natural images and long PDF documents. Modern VLMs accept dynamic resolutions and allocate vision tokens roughly in proportion to pixel area, directly or through tiling [2, 3, 7, 22, 40], which makes the rendering budget a meaningful and controllable cost. We therefore report fidelity on synthetic needles and on naturally occurring identifiers separately from collateral damage, and we measure cost in provider-billed tokens rather than in pixels.

# 3. Method

**Overview.** Given a long document, an optical compressor renders it into image pages that a VLM reads in place of the text. Our goal is to make the strings that must be copied exactly survive this rendering, without spending more vision tokens and without access to the model. Figure 2 shows the pipeline. DETECT reads the text and marks verbatim-hazardous strings, and ALLOCATE, our contribution, renders those strings large and the remaining prose small, so that the image costs no more than uniform rendering would. The other two steps follow prior work on failure-aware optical compression [1]: ANCHOR appends verbatim text copies of the flagged strings, here paid for by the vision tokens the allocation saved, and ROUTE falls back to plain text when the anchors would be too expensive.

## 3.1. Problem formulation

A document $x = (w_1, \ldots, w_n)$ is a sequence of whitespace-separated words. A renderer $R$ maps $x$ and a size assignment $\sigma : \{1, \ldots, n\} \to \mathbb{N}$ (glyph height in pixels for every word) to an image $I = R(x; \sigma)$ of fixed width, which is split into pages for the VLM. To compare renderings, we need their cost in the unit providers bill. Following the patch-based accounting of dynamic-resolution VLMs [40], we define the vision-token cost of an image with pages of width $w_p$ and height $h_p$ as

$$v(I) = \sum_{p} \left\lceil \frac{w_p}{28} \right\rceil \left\lceil \frac{h_p}{28} \right\rceil , \tag{1}$$

where each vision token covers a $28 \times 28$-pixel area (14-pixel patches merged $2 \times 2$ in Qwen2-VL). Equation (1) is used only to enforce budgets; all costs we report are the prompt tokens billed by each provider. The *budget* is the cost of uniform rendering at a base size $s_0$, $B = v\big(R(x; \sigma \equiv s_0)\big)$, so every method we compare uses the same number of vision tokens as the uniform baseline, up to page rounding. Let $\mathcal{H}(x) \subseteq \{1, \ldots, n\}$ be the indices of hazardous words, those whose value is needed verbatim downstream. We seek a size assignment that makes every $w_i$, $i \in \mathcal{H}(x)$, as legible as possible subject to $v(I) \leq B$, without deleting or reordering any character.

## 3.2. DETECT: text-side hazard detection

The detector approximates $\mathcal{H}(x)$ without labels, following the regex-plus-windowed-entropy design of [1]. Its premise is that a string is hazardous when it is *high-entropy*: a misread character yields another valid-looking string, and the language model cannot repair it from context. After stripping surrounding punctuation, we flag a word if it matches one of five patterns: a hexadecimal run of at least 8 characters, a run of at least 5 digits, a path containing a digit or a file extension, an IP address, or a digest prefix. We also flag

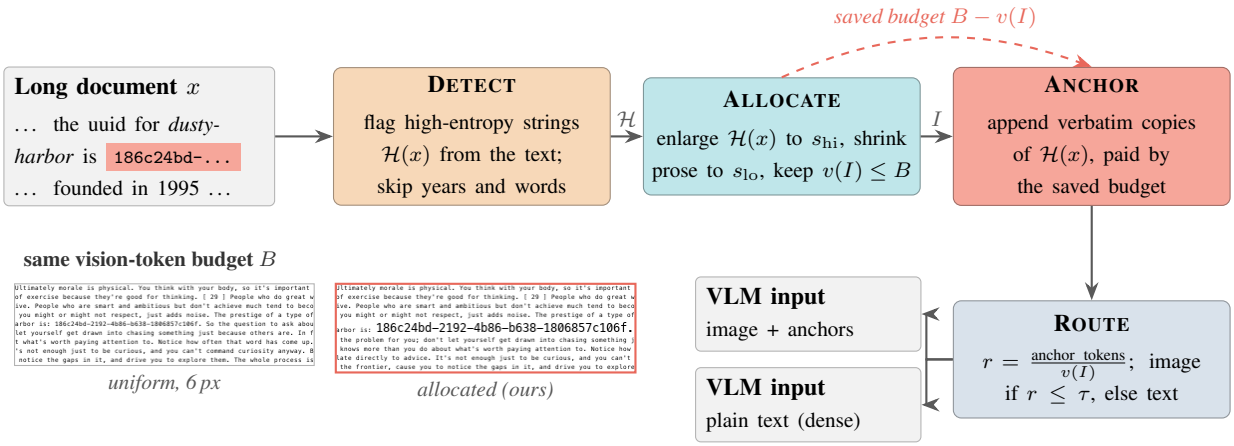



Figure 2. **Hazard-aware rendering.** Our contribution is ALLOCATE. DETECT marks verbatim-risky strings in the text before rendering; ALLOCATE enlarges only those strings and shrinks prose so that the image stays within the vision-token budget $B$ of uniform rendering (up to page rounding); ANCHOR and ROUTE, which follow prior work [1], spend the saved budget on verbatim copies of the flagged strings and send documents whose anchors would be too expensive to plain text. The thumbnails show the same 32K-character document rendered uniformly and by our allocation. Every step is training-free and needs no access to the model.

words that mix letters and digits and either have at least 20 identifier characters or contain a 16-character window with Shannon entropy of at least 3.5 bits. Two choices encode what *not* to protect. Four-digit numbers are excluded because in natural text they are mostly years, which are low-entropy and cheap to read. The length and entropy rules require mixed letter–digit tokens so that long English words are not flagged. These exclusions are our changes; a preliminary detector without them flagged on average 85–112 words per LongBench document, most of them years, and shrank prose in documents that did not need it (Sec. D).

When the detector flags nothing, the rendering is identical to the uniform baseline, so a document whose hazards all go undetected loses only the potential gain (Sec. 5.7). A missed hazard in a document where *other* strings are flagged is, however, shrunk together with the prose.

### 3.3. ALLOCATE: budget-constrained glyph allocation

Given $\mathcal{H}(x)$, we restrict size assignments to two levels: prose words at $s_{\text{lo}}$ and flagged words at $s_{\text{hi}}$. Two levels keep the layout simple; a locally varying alternative that shrinks only prose near flagged strings brought no further gain (Sec. D). Prose is wrapped at character boundaries like the uniform baseline, so it consumes no extra space, while each flagged word is kept on one line so it is never split. We choose the sizes by

$$(s^*_{\text{hi}}, s^*_{\text{lo}}) = \operatorname*{lexmax}_{\substack{s_{\text{hi}} \in [s_0,\, 12] \\ s_{\text{lo}} \in [5,\, \min(s_{\text{hi}}, s_0)]}} (s_{\text{hi}}, s_{\text{lo}}) \quad \text{s.t. } v\big(R(x; \sigma_{s_{\text{lo}}, s_{\text{hi}}})\big) \le B, \tag{2}$$

where $\sigma_{s_{\text{lo}}, s_{\text{hi}}}$ assigns $s_{\text{hi}}$ to words in $\mathcal{H}(x)$ and $s_{\text{lo}}$ to all others, and lexmax first maximizes $s_{\text{hi}}$ and then $s_{\text{lo}}$. The order expresses the design choice that the marginal pixel is worth more on a hash than on prose: verbatim accuracy is limited by the least legible character of a string, whereas prose comprehension degrades gracefully. The lower bound of 5 px keeps prose within the range VLMs still read, and $s_{\text{lo}} \le s_0$ ensures prose is never enlarged. If no pair with $s_{\text{hi}} > s_{\text{lo}}$ satisfies the constraint, prose-only lines lose their single pixel of extra line spacing and the search is repeated (in the implementation, $s_{\text{hi}}$ is searched over $[6, 12]$); if $\mathcal{H}(x) = \emptyset$, the result is identical to uniform rendering. At $s_0 = 5$ px, prose is already at its minimum size, so this removed line spacing is what makes room for the enlarged strings; at $s_0 = 6$ px, prose shrinks to 5 px when needed and otherwise stays at 6 px. Budget computation plus allocation takes about 1 s for 32K and 4 s for 128K characters on one CPU core (unoptimized Python). Figure 1 (middle) shows a typical outcome: the UUID is drawn at 12 px while prose stays at 6 px.

### 3.4. ANCHOR and ROUTE

When prose can shrink ($s_0 = 6$ px), the allocated image usually ends *below* the budget, because the search stops at the largest feasible $s_{\text{hi}}$ rather than filling every remaining row; in our long-context runs it used 13–27% fewer billed tokens than uniform rendering. Following [1], we spend these savings on *anchors*: after the question, the flagged strings are appended verbatim as text, under a line stating that they are copied exactly from the image. Anchors carry no key, but when a document contains only one or a few flagged strings, the anchor list is short and may consist of

the answer alone, so a reader can answer without reading the image (Sec. 5.5). Anchor cost grows linearly with the number of flagged strings, so before any model call we compute the ratio

$$r(x) = \frac{\sum_{i \in \mathcal{H}(x)} c(w_i)}{v(I)}, \qquad c(w) = \frac{|w|}{2} + 3, \qquad (3)$$

where $c(w)$ estimates the text tokens of one anchor and $I$ is the allocated image, and send the document as plain text when $r(x) > \tau$. This replaces the density test of [1] with one that accounts for the allocated image; we use $\tau = 0.3$, the median of per-reader two-fold cross-validation on a synthetic density sweep (Sec. E).

## 4. Experimental Setup

**Readers.** We treat every VLM as a black box and query it with one stateless request per item at temperature 0 (0.01 for MiniMax); a request whose output cannot be parsed is resent up to three times. The six readers span vendors, scales, and open weights: Claude Sonnet 5.5, Gemini 3.5 Flash-Lite, GPT-6 Luna (minimal reasoning effort), Qwen3-VL-235B-A22B [2], Llama 4 Maverick, and MiniMax-M3. All means and pooled tests use the five readers other than Claude, which was not run on every arm and is reported per reader only. Costs are the prompt tokens billed by each provider, which include the image, the question, and any appended copy; MiniMax does not report usage and is excluded from cost averages only. Empty responses and transport errors (fewer than 100 of roughly 35,000 calls in all experiments reported here, about 0.25%) are dropped; any other unparseable output, including refusals and degenerate repetition, counts as wrong.

**Tasks.** Verbatim fidelity is measured with RULER needle-in-a-haystack tasks [16], re-implemented with the official templates and Paul Graham essays as the haystack: *single* (one UUID value), *multi-key* (four keys, one queried, numeric values), and *multi-value* (four numeric values for one key). Contexts have 16K characters (Sec. 5.1), 32K–128K characters split into up to three pages (Sec. 5.2), or 32K characters at milder budgets (Sec. 5.3). Collateral damage on ordinary comprehension is measured on LongBench [4] MultiFieldQA-en and HotpotQA (from the length-balanced LongBench-E split), with non-ASCII characters replaced by "?", using contexts of at most 40K characters so that one image holds a document; these samples were drawn after the detector was finalized. We use $N = 25$ items per task and reader for RULER at 16K and 32K characters, $N = 20$ for the 32K–128K length study, and $N = 50$ for LongBench.

**Arms.** All image arms share the renderer of Sec. 3, a 1008-pixel page width, and the budget $B$ of uniform rendering at $s_0 \in \{5, 6\}$ px (≈32 and 21 characters per vision token on our long documents). **A** is uniform rendering. **U** searches font family (monospace, Arial, Arial Narrow) and size for the largest uniform text that fits $B$, a training-free stand-in for global configuration search [8]. **Z** renders at 10 px and downsamples the image to $B$, the resolution-compression view of optical encoders [41]. **R** applies our allocation to randomly chosen ordinary words, matched to within 10% of the total length of the flagged strings, and **B** draws S's layout without enlarging anything but boxes each flagged string, a salience-only control. **S** is our allocation (Sec. 3.3). **K** and **X** append anchors [1] to A and to S, and **T** gives the plain text. Prompts follow the task templates, with the context replaced by the image; RULER answers are requested as JSON.

**Metrics and statistics.** RULER is scored by recall of the gold values (case-insensitive containment, all values), LongBench by token-level F1, as in the original benchmarks. Because recall does not penalize extra answers, we also report a precision-aware F1 over the returned values. For each reader on each task at each setting (a *cell*) we compute the mean paired difference over items with a 95% bootstrap confidence interval (10,000 resamples). Pooled effects average over cells, with 95% CIs from a bootstrap that resamples both readers and items, and we count significant cells after Holm correction within each comparison (Tab. 2).

## 5. Results

### 5.1. Equal-budget comparison

Table 1 compares all image arms at the same vision-token budget on 16K-character RULER documents. Averaged over five readers, uniform rendering recovers 8% of needles at the denser budget and 22% at the lighter one; Claude Sonnet 5.5 (0.57) and Gemini (0.88) are the only readers above 0.11 at 6 px. Allocation (S) raises the mean to 0.64 and 0.65, a pooled gain of 0.55 and 0.43 whose confidence intervals exclude zero by a wide margin (Tab. 2); after Holm correction it is significant in 14 and 11 of 15 cells and never significantly negative. The two stronger uniform baselines barely help: searching font family and size (U) reaches 0.12 and 0.21, and rendering large then downsampling (Z) 0.16 and 0.19, plausibly because both spread legibility evenly over all characters. At 6 px, S is also 8% cheaper than uniform rendering (842 vs. 920 billed tokens), because prose shrinks more than the enlarged strings grow. The weaker readers gain most (at 6 px Llama 4 Maverick from 0.00 to 0.59, GPT from 0.11 to 0.65), while Gemini, already at 0.88, gains 0.12. Under the precision-aware F1, which penalizes returning several candidate values, the gains are smaller (0.46 and 0.34 pooled) but remain significant in at

Table 1. **Equal-budget comparison on RULER needle retrieval (16K characters).** Recall per reader averaged over three NIAH tasks ($N$=25 each), at two budgets (uniform 5 px ≈ 32 and 6 px ≈ 21 characters per vision token). A, U, Z, R, and S are images with the same vision-token budget: uniform, best uniform font/size, render-then-downsample, *random* ordinary words enlarged by our allocation (matched in total length), and flagged strings enlarged (ours). K and X append anchors [1] to A and S. **Bold**: best within each group. Means, F1 (precision-aware: extra returned values count against the reader), and billed tokens are over the five readers other than Claude‡, which was not run on R and X. Pooled over tasks and readers, S−A is +0.55 [0.48, 0.62] at 5 px and +0.43 [0.27, 0.56] at 6 px (95% CI, readers and items resampled; Tab. 2). On these tasks the anchors contain the answer (Sec. 5.5), so K and X measure copying rather than reading.

| | 5 px budget (≈32 chars/vtok) | | | | | | | 6 px budget (≈21 chars/vtok) | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | equal-budget image | | | | | + anchors | | equal-budget image | | | | | + anchors | |
| Reader | A | U | Z | R | S (ours) | K | X | A | U | Z | R | S (ours) | K | X |
| Claude Sonnet 5.5‡ | 0.14 | 0.22 | 0.19 | – | **0.79** | **0.95** | – | 0.57 | 0.64 | 0.46 | – | **0.92** | **1.00** | – |
| Gemini 3.5 Flash-Lite | 0.41 | 0.57 | 0.79 | 0.48 | **1.00** | **1.00** | **1.00** | 0.88 | 0.78 | 0.84 | 0.46 | **1.00** | 1.00 | **1.00** |
| GPT-6 Luna | 0.02 | 0.03 | 0.02 | 0.02 | **0.50** | 0.91 | **0.94** | 0.11 | 0.16 | 0.08 | 0.02 | **0.65** | 0.82 | **0.98** |
| Qwen3-VL-235B | 0.00 | 0.00 | 0.00 | 0.00 | **0.55** | **1.00** | **1.00** | 0.07 | 0.08 | 0.03 | 0.00 | **0.60** | **1.00** | **1.00** |
| Llama 4 Maverick | 0.00 | 0.00 | 0.00 | 0.00 | **0.58** | **0.97** | **0.97** | 0.00 | 0.01 | 0.00 | 0.00 | **0.59** | 0.90 | **1.00** |
| MiniMax-M3 | 0.00 | 0.00 | 0.00 | 0.00 | **0.57** | **1.00** | 0.98 | 0.02 | 0.01 | 0.00 | 0.00 | **0.42** | 0.97 | **1.00** |
| *Mean recall* | 0.08 | 0.12 | 0.16 | 0.10 | **0.64** | 0.97 | **0.98** | 0.22 | 0.21 | 0.19 | 0.10 | **0.65** | 0.94 | **1.00** |
| *Mean F1* | 0.09 | 0.12 | 0.16 | 0.10 | **0.55** | 0.80 | **0.80** | 0.23 | 0.21 | 0.19 | 0.10 | **0.56** | 0.79 | **0.83** |
| *Billed tokens* | 815 | 811 | 804 | 814 | 818 | 860 | 857 | 920 | 902 | 894 | 844 | 842 | 968 | 881 |

least 9 of 15 cells.

**Targeting, enlargement, or salience?** Two controls separate the ingredients of S. Arm R runs the same allocation on randomly chosen ordinary words, matched in total length, so its sizes and cost are close to those of S while the needles shrink with the prose; it recovers 0.10 of the needles at both budgets, and S exceeds it by 0.54 and 0.56. Arm B keeps every glyph at the uniform size but draws a box around each flagged string, which makes the needle easy to *find* but not easier to *read*; it is slightly worse than uniform rendering (−0.05, CI upper bound −0.001), perhaps because the box line touches the small glyphs, and 0.53 below S (Tab. 2). The gain therefore comes from making the targeted strings legible, not from layout changes or from pointing at the needle.

## 5.2. Scaling with context length

Optical compression is most attractive for long documents, so Fig. 3 extends the comparison to 128K characters. Under uniform rendering, mean recall declines from 0.12 at 32K to 0.08 at 64K and 0.07 at 128K, consistent with reports that compressed reading loses retrieval ability as context grows [52]. Allocation stays between 0.60 and 0.56, a pooled gain of 0.48 [0.43, 0.53] with 35 of 45 cells significant after Holm correction. It also uses 13%, 23%, and 27% fewer billed tokens than uniform rendering at the three lengths and, on the four readers with billed costs for both,

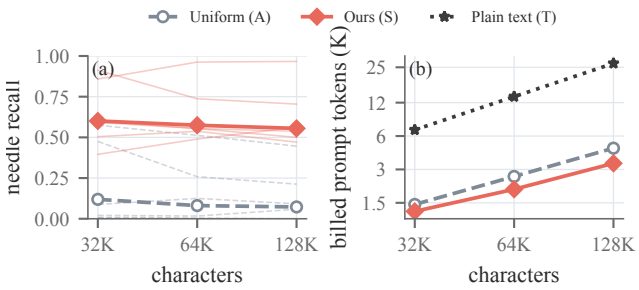



Figure 3. **Context length.** RULER at 32K–128K characters (6 px budget, up to three image pages, $N$=20 per task). (a) Needle recall; thin lines are individual readers (including Claude), thick lines the mean over five readers. (b) Billed prompt tokens averaged over the four of these readers that report usage; note the log scale.

5.4×, 6.8×, and 8.0× fewer than the plain text, which every reader answers with recall of at least 0.99. Individual readers differ widely (0.47 to 0.97 at 128K for S; Tab. 5).

## 5.3. Milder budgets

Our main budgets put most readers near the floor of uniform rendering, where gains are easiest to obtain. We therefore repeated the single and multi-key tasks at 32K characters with base sizes of 8 and 10 px, which cost 1.5 and 2.3 times as many vision tokens as 6 px. Uniform rendering improves with size (multi-key: 0.26 at 8 px, 0.60 at 10 px), and the

Table 2. **Pooled effects** over the five non-Claude readers (recall unless noted). CI: 95% bootstrap resampling readers and items. Sig.: cells (reader × task) significantly positive after Holm correction within each row; neg.: significantly negative. All X−K comparisons we ran are listed.

| Comparison | Δ | 95% CI | sig. | neg. |
|---|---|---|---|---|
| *Targeting vs. uniform and alternatives (RULER)* | | | | |
| S−A, 16K, 5 px | +0.55 | [+0.48, +0.62] | 14/15 | 0 |
| S−A, 16K, 6 px | +0.43 | [+0.27, +0.56] | 11/15 | 0 |
| S−A, 32K–128K, 6 px | +0.48 | [+0.43, +0.53] | 35/45 | 0 |
| S−U (font/size search), 6 px | +0.44 | [+0.31, +0.55] | 11/15 | 0 |
| S−R (random words), 6 px | +0.56 | [+0.47, +0.64] | 12/15 | 0 |
| B−A (box, no enlargement), 6 px | -0.05 | [-0.13, 0.00] | 0/15 | 1 |
| S−B, 6 px | +0.53 | [+0.41, +0.64] | 14/15 | 0 |
| *Milder budgets (32K)* | | | | |
| S−A, 8 px | +0.38 | [+0.28, +0.46] | 8/10 | 0 |
| S−U, 8 px | +0.26 | [+0.14, +0.40] | 3/10 | 0 |
| S−A, 10 px | +0.20 | [+0.11, +0.30] | 4/10 | 0 |
| S−U, 10 px | +0.04 | [-0.04, +0.12] | 1/10 | 1 |
| *Natural strings (LongBench documents)* | | | | |
| S−A, flagged targets | +0.25 | [+0.12, +0.38] | 4/5 | 0 |
| S−A, detector-independent | -0.01 | [-0.11, +0.08] | 0/5 | 0 |
| *Ordinary comprehension (LongBench F1)* | | | | |
| S−A | +0.02 | [-0.02, +0.06] | 0/10 | 0 |
| X−A | +0.02 | [-0.03, +0.07] | 0/10 | 0 |
| X−T (plain text) | -0.21 | [-0.29, -0.13] | 0/10 | 9 |
| *Anchors: X−K (F1 on RULER, recall on natural strings)* | | | | |
| RULER 16K, 5 px | 0.00 | [-0.02, +0.03] | 0/15 | 0 |
| RULER 16K, 6 px | +0.04 | [0.00, +0.08] | 1/15 | 0 |
| RULER 32K, 8 px | 0.00 | [-0.04, +0.04] | 0/10 | 0 |
| RULER 32K, 10 px | 0.00 | [-0.05, +0.06] | 0/10 | 0 |
| density $k \leq 16$, 32K | -0.01 | [-0.07, +0.04] | 1/15 | 0 |
| multi-key 64K–128K | -0.01 | [-0.05, +0.03] | 0/10 | 0 |
| natural, flagged, 5 px | +0.04 | [-0.03, +0.11] | 0/5 | 0 |
| natural, flagged, 6 px | +0.02 | [-0.06, +0.10] | 0/5 | 0 |
| natural, flagged, 7 px | +0.05 | [-0.06, +0.16] | 1/5 | 0 |
| natural, detector-indep. | -0.08 | [-0.17, +0.01] | 0/5 | 1 |

gain of S shrinks accordingly, to 0.38 at 8 px and 0.20 at 10 px (Tab. 2). The font/size search catches up: S beats U by 0.26 at 8 px but only by 0.04 at 10 px, which is not significant. Single UUIDs remain hard even here (A 0.20, S 0.44 at 10 px), but allocation is most useful where the budget is tight, which is also where optical compression saves the most.

## 5.4. Naturally occurring strings

RULER needles are synthetic, and our detector was designed with such strings in mind. We built two sets of 60 questions from LongBench documents of up to 40K characters (ten each from six tasks), each asking for the exact string that follows a unique six-word phrase (Sec. F). In the first set, targets are strings the detector flags, such as report and grant numbers, ISBNs, citation keys, and URLs. Here

Table 3. **Naturally occurring strings.** Exact-copy recall on LongBench documents (≤40K characters, 6 px budget, $N$=60 each). *Flagged*: targets the detector flags (report and grant numbers, ISBNs, URLs, . . . ). *Detector-independent*: any digit-bearing token, sampled without the detector (1/60 flagged; mostly short numbers and versions). $^{*}/^{\dagger}$: paired 95% CI vs. A above/below zero. Billed tokens: four readers that report usage.

| | Flagged targets | | | | | Detector-independent targets | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Reader | A | S | K | X | T | A | S | K | X | T |
| Gemini 3.5 Flash-Lite | 0.72 | 0.80 | 0.87$^{*}$ | 0.90$^{*}$ | 0.92 | 0.85 | 0.87 | 0.82 | 0.73$^{\dagger}$ | 1.00 |
| GPT-6 Luna | 0.25 | 0.55$^{*}$ | 0.85$^{*}$ | 0.82$^{*}$ | 1.00 | 0.57 | 0.52 | 0.62 | 0.40$^{\dagger}$ | 0.98 |
| Qwen3-VL-235B | 0.12 | 0.48$^{*}$ | 0.77$^{*}$ | 0.78$^{*}$ | 0.98 | 0.33 | 0.43 | 0.32 | 0.28 | 0.98 |
| Llama 4 Maverick | 0.05 | 0.20$^{*}$ | 0.55$^{*}$ | 0.62$^{*}$ | 0.97 | 0.25 | 0.25 | 0.23 | 0.23 | 0.88 |
| MiniMax-M3 | 0.03 | 0.39$^{*}$ | 0.65$^{*}$ | 0.67$^{*}$ | 0.98 | 0.27 | 0.15 | 0.22 | 0.17 | 0.93 |
| *Mean* | 0.23 | 0.48 | 0.74 | 0.76 | 0.97 | 0.45 | 0.44 | 0.44 | 0.36 | 0.96 |
| *Billed tokens* | 1181 | 1035 | 1267 | 1128 | 5324 | 1111 | 1067 | 1225 | 1173 | 4942 |

allocation doubles mean recall (0.23 to 0.48, significant for four of five readers) at 12% fewer billed tokens than uniform rendering (Tab. 3). In the second set, targets are any digit-bearing tokens of at least four characters, sampled without the detector; the detector flags only one of the 60, because most are short numbers, percentages, and version strings such as “25,000” or “2.0.13”. On these, every image arm is poor (0.36–0.45 against 0.96 for text), and allocation neither helps nor hurts (−0.01 [−0.11, 0.08]). Shrinking prose to make room for flagged strings thus did not measurably damage the strings the detector misses, but nothing in the image protects them either.

## 5.5. Anchors on the allocated image

Anchors [1] copy the flagged strings into the prompt, so their value depends on how many strings compete for the answer. On the single-needle and multi-value tasks of Tab. 1, the only flagged strings in the essay haystack are the needles, so the anchor list *is* the answer (1.0 anchor per single-needle document, always the gold value), and K and X reach 0.90–1.00. On multi-key, the list adds three distractor values; recall stays at 0.94–1.00, but readers often return the whole list, and the precision-aware F1 drops to 0.46–0.52. These numbers measure copying rather than reading, so we compare K and X where the list does not reveal the answer.

On the flagged natural identifiers, documents carry 5.3 distinct anchors on average, and choosing one at random would score 0.44. Readers do better (K 0.74, X 0.76) and almost always return a single value, so here the anchors are used, not dumped. Repeating this set at base sizes of 5, 6, and 7 px gives a recall–cost frontier for each arm: X reaches 0.75, 0.76, and 0.81 at 1017, 1128, and 1306 billed tokens, and K reaches 0.71, 0.74, and 0.77 at 1014, 1267, and 1345. X thus lies on or above K’s frontier, by 0.02–0.05 at equal or lower cost (equal at 5 px, where prose cannot shrink), although no single difference is significant (Tab. 2). At 64K

Table 4. **Ordinary comprehension.** LongBench QA F1 on held-out samples ($N$=50 per task, contexts ≤40K characters, 6 px budget, five readers). $^{*}/^{\dagger}$: paired 95% CI vs. uniform rendering (A) above/below zero. Relative to A, X has no significantly negative cell and S one; every image arm trails plain text T (X−T significantly negative in 9 of 10 cells). Billed tokens: four readers that report usage.

| | MultiFieldQA-en | | | | | HotpotQA | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Reader | A | S | K | X | T | A | S | K | X | T |
| Gemini 3.5 Flash-Lite | 0.47 | 0.47 | 0.53 | 0.47 | 0.63 | 0.59 | 0.62 | 0.58 | 0.57 | 0.78 |
| GPT-6 Luna | 0.36 | 0.39 | 0.37 | 0.36 | 0.56 | 0.68 | 0.73 | 0.68 | 0.70 | 0.72 |
| Qwen3-VL-235B | 0.27 | 0.26 | 0.31 | 0.26 | 0.59 | 0.66 | 0.61$^{\dagger}$ | 0.65 | 0.62 | 0.77 |
| Llama 4 Maverick | 0.28 | 0.25 | 0.34 | 0.29 | 0.59 | 0.45 | 0.50 | 0.51$^{*}$ | 0.50 | 0.68 |
| MiniMax-M3 | 0.23 | 0.26 | 0.25 | 0.25 | 0.56 | 0.29 | 0.34 | 0.41$^{*}$ | 0.42$^{*}$ | 0.69 |
| *Mean* | 0.32 | 0.33 | 0.36 | 0.33 | 0.58 | 0.53 | 0.56 | 0.57 | 0.56 | 0.73 |
| *Billed tokens* | 1141 | 1056 | 1440 | 1355 | 5342 | 1200 | 1114 | 1239 | 1153 | 5617 |

and 128K characters, X costs 31% and 29% fewer billed tokens than K because allocation shrinks the image, but on multi-key the precision-aware F1 is the same for S, K, and X (0.36–0.40), so there the anchors add nothing over allocation alone. When the target is *not* flagged, anchors can mislead: on the detector-independent set, X scores 0.36 against 0.44 for S and K (−0.08 vs. K, CI [−0.17, 0.01]), consistent with readers copying a plausible anchor instead of reading the image. On a synthetic sweep of hazard density (Sec. E), anchors on either image become expensive and ineffective beyond about 16 flagged strings per 32K characters, which is where routing to text [1] takes over.

### 5.6. Effect on ordinary comprehension

Shrinking prose could hurt questions that have nothing to do with identifiers. Table 4 tests this on held-out LongBench documents of up to 40K characters. Relative to uniform rendering, F1 is unchanged on MultiFieldQA (0.32 for A, 0.33 for S and X) and slightly higher on HotpotQA (0.53 for A, 0.56 for S and X); X has no significantly negative cell, and S has one (Qwen3-VL on HotpotQA, −0.06). Pooled over readers and tasks, S−A is +0.02 [−0.02, 0.06] and X−A +0.02 [−0.03, 0.07] (Tab. 2), which rules out losses larger than about 0.03 F1. Relative to plain text, however, every image arm loses: T reaches 0.58 and 0.73, and X trails it by 0.15–0.33 F1 in 9 of 10 cells, at a fifth to a quarter of the text cost. Only GPT-6 Luna reads the HotpotQA images nearly as well as text. Allocation therefore removes a failure specific to identifiers without making optical compression a lossless substitute for text; whether the remaining comprehension gap is acceptable depends on the reader and the application.

### 5.7. Analysis

**How much does the detector matter?** Replacing the regex detector by an oracle that flags exactly the needle values changes nothing when the needles are of a type the detector recognizes (seven-digit numbers: 0.77 vs. 0.78 for S; Tab. 6). When they are not (four-digit numbers, which the detector deliberately ignores), S falls back to the uniform rendering in 24 of 25 documents and matches its recall (0.33 vs. 0.32), while the oracle would reach 0.82. Together with the detector-independent set of Sec. 5.4, this suggests that missed hazards cost the gain but, in our tests, not more.

**Where do real documents fall?** Because the routing ratio is computed from text alone, we profiled 32K-character chunks of 13 LongBench tasks, essays, Python source, system logs, and package manifests without querying any model (Sec. G). Prose-dominated documents are routed to images 95–100% of the time at an estimated 0.19–0.27× the plain-text cost and technical prose (MultiFieldQA) 75% of the time, while code is mostly routed to text and logs and manifests always are. Hazard-aware rendering is thus a tool for long documents that *contain* identifiers, not for documents that *consist* of them.

## 6. Limitations and Conclusion

**Limitations.** The gains are largest at dense budgets where uniform rendering fails; at 10 px a font search does as well. Optical compression, with or without allocation, trails plain text on ordinary QA by up to 0.33 F1 for weaker readers, and the router decides from identifier cost alone, not from this gap. Most fidelity tests use synthetic needles with sparse distractors; our natural sets are small (60 automatically constructed questions each), the detector protects only high-entropy strings, and anchors mislead readers when the target is not flagged. We evaluate black-box readers only, through public APIs whose versions and image preprocessing may change, and cannot say how enlarged glyphs change internal attention; trained optical compressors [8, 41] and token pruning on open models are untested. Budgets follow one patch-based accounting rule (Eq. (1)), and providers tile images differently: billed savings hold in most settings, with exceptions at short lengths (Gemini up to 32K characters, Qwen at 16K and 5 px, and Llama, whose billing ignores image size). We use ASCII English only, and we do not measure latency, prompt caching [15], or the fact that text inside images bypasses text-side safety filters.

**Conclusion.** Optical compression decides how many pixels a document receives but not where they go. Because the renderer reads the text before drawing it, it can give the strings that must be copied exactly the pixels they need, within the budget of uniform rendering and without access to the model. Across five readers, this raises needle recall by about half the scale at dense budgets, doubles the recall of flagged natural identifiers, and makes the anchors of prior

work cheaper where prose can shrink, while leaving the gap to plain text on ordinary comprehension in place.

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## LLM Usage Statement

Large language models play two roles in this work. First, they are the *objects of study*: the six vision-language models listed in Sec. 4 and the additional Claude reader of Sec. H were queried as black-box readers, and all reported numbers are computed by our scripts from their logged outputs. Second, the authors used an LLM-based coding assistant (Claude) to help implement the rendering and evaluation code, to generate figures and tables from the logged results, and to draft and edit the text of this paper. The assistant also proposed experiments and analyses, which the authors reviewed and approved; the authors take responsibility for all claims. Every number in the paper is produced programmatically from the experiment logs, and every reference was checked against its primary source. No LLM was used to generate or label evaluation data beyond the readers' answers being evaluated.

# A. Implementation Details

**Renderer.** Pages are 1008 pixels wide (36 patches) with a 4-pixel margin and are rendered in grayscale with the Menlo monospace font through PIL. Prose is wrapped at character boundaries; a flagged word is placed whole on the next line when it does not fit. Each line is as tall as its largest glyph plus one pixel of leading. Documents taller than 2016 pixels are split into pages at blank pixel rows, so no line is cut; all pages are sent in one request in reading order. The budget of Eq. (1) is enforced on the unsplit image; splitting can add one patch row per page, which made 10 of the 180 reallocated long-context images 1.2% more expensive than their uniform counterparts. All reported costs are billed tokens and include this effect.

**Detector.** After stripping the characters `.,;:()[]"'` from both ends, a word is flagged if it matches `[0-9a-f]{8,}`, `\d{5,}`, `/\S*\d\S*`, `\d+\.\d+\.\d+\.\d+`, `\S+/\S+\.\w{1,5}`, or `sha256=`, or if it contains at least one letter and one digit and either contains `[A-Za-z0-9_\-]{20,}` or is at least 16 characters long with a 16-character window whose character entropy is at least 3.5 bits.

**Baselines.** U tries Menlo, Arial, and Arial Narrow at sizes 14 down to 5 px and keeps the largest size that fits the budget (ties broken by font order); at the 5 px and 6 px budgets it selects 6 px Arial and 8 px Arial Narrow for most documents. Z renders at 10 px and shrinks the image by repeated factors of 0.98 with Lanczos resampling until it fits the budget (final scale 0.52–0.64). For RULER and the natural-identifier questions, K and X append, after the question, the line "Exact strings copied verbatim from the text block in the image:" followed by one flagged string per line; for LongBench, the anchors are inserted after the image placeholder and before the question, under "Exact strings copied verbatim from the text in the image:".

**Prompts and scoring.** RULER prompts follow the official template with the context replaced by "[The text is rendered in the image above.]", add "Copy each one exactly, character for character.", and ask for a JSON list of values; a gold value counts as recalled if it appears (case-insensitively) in any returned string. LongBench prompts are the official ones for each task; the first output line is scored with the official QA F1. For the plain-text arm the context is inserted verbatim into the same template.

**Router threshold.** $\tau$ is chosen per reader by two-fold cross-validation on the density sweep (folds by item parity), maximizing recall minus 0.5 times the cost relative to plain text over $\tau \in \{0, 0.05, 0.1, 0.2, 0.3, 0.5, 0.75, 1, 1.5, 2, 4, \infty\}$. Selected values were 0.1–0.5 with median 0.3, which we use for all other experiments. The frontier in Fig. 5c sweeps $\tau$ and the per-density figures use the median $\tau$; both are therefore in-sample, unlike the cross-validated aggregate.

# B. Per-Reader Results

Tables 5 and 8 give per-reader numbers for Figs. 3 and 5.

Table 5. Per-reader recall for the length study (Fig. 3).

| | 32K | | 64K | | 128K | |
|---|---|---|---|---|---|---|
| Reader | A | S | A | S | A | S |
| Claude Sonnet 5.5 | 0.57 | **0.86** | 0.51 | **0.96** | 0.45 | **0.97** |
| Gemini 3.5 Flash-Lite | 0.47 | **0.91** | 0.26 | **0.74** | 0.21 | **0.70** |
| GPT-6 Luna | 0.09 | **0.59** | 0.12 | **0.55** | 0.09 | **0.50** |
| Qwen3-VL-235B | 0.02 | **0.60** | 0.02 | **0.55** | 0.06 | **0.54** |
| Llama 4 Maverick | 0.01 | **0.50** | 0.00 | **0.54** | 0.00 | **0.47** |
| MiniMax-M3 | 0.00 | **0.40** | 0.00 | **0.49** | 0.00 | **0.56** |
| *Mean (5 readers, excl. Claude)* | 0.12 | 0.60 | 0.08 | 0.57 | 0.07 | 0.56 |

Table 6. **Detector ablation.** RULER multi-value recall (16K characters, 6 px budget, $N$=25, mean of 5 readers; Claude was not run in this ablation). Seven-digit values are flagged by the regex detector; four-digit values are deliberately not (they collide with years), so the regex arms fall back to uniform rendering.

| Arm | 7-digit values (detected) | 4-digit values (missed) |
|---|---|---|
| Uniform (A) | 0.22 | 0.32 |
| Ours, regex detector (S) | 0.77 | 0.33 |
| Ours, oracle detector | 0.78 | 0.82 |
| Ours + anchors, regex (X) | 0.99 | 0.32 |
| Ours + anchors, oracle | 0.99 | 0.99 |

# C. Short Documents: Mechanism Study

Before the long-context experiments, we studied 1.6K-character documents that fit one small image, with six identifiers and three one-word prose facts per document ("The release codename is . . . "), both on synthetic text and on real package manifests whose lines contain file paths and 43-character base64 hashes. This setting isolates the allocation mechanism but is not a deployment target: for such short inputs an image is often not cheaper than the text.

**Reallocation helps every reader; prose can pay for it.** Table 7 reports word-wrapped reallocation against uniform rendering. Identifier exact-match improves significantly in all 24 reader–condition cells. Prose probes are unaffected

on synthetic text except for Llama at 14 characters per vision token (−0.39). On real manifests, weaker readers lose up to 0.38 of prose accuracy, because whole manifest lines are enlarged and prose shrinks from 6 to 5 px. On long documents, where hazards occupy a small fraction of the text, this cost does not appear (Tab. 4).

**The gain depends on density and reader.** Figure 4 sweeps four densities from 7.8 to 21.1 characters per vision token on synthetic documents. The gain follows an inverted U whose peak shifts with reader strength. The strongest readers gain most at the highest density (Claude +0.32 and Gemini +0.39 at 21 characters per vision token), where they first start to fail. Weaker readers gain most at moderate densities (+0.48 to +0.78 at 11–14).

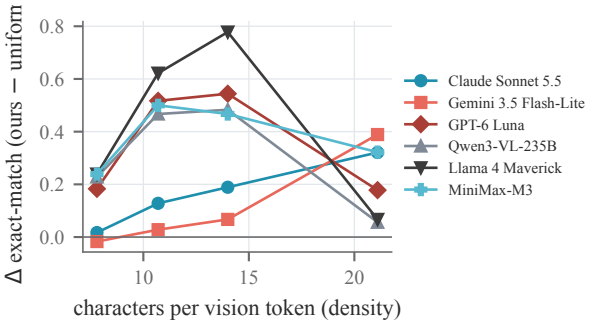



Figure 4. Gain in identifier exact-match of reallocation over uniform rendering versus rendering density (short synthetic documents, $N$=30 per point).

**Readers do not know when they misread.** Allowing an "[unreadable]" answer did not consistently reduce silent errors: MiniMax-M3 used it for at most 6% of identifiers, and its rate of wrong-but-well-formed answers fell in one of four conditions (uniform, 21 characters per vision token: 0.40 to 0.31) and rose in the other three. This supports preventing errors at render time rather than relying on the reader's self-assessment.

**Agent-based readers can leak across items.** In a pilot, Claude readers accessed through an agent interface read several images in one session. Among their wrong prose answers, 50–100% were names that appeared in *other* images of the same session, versus 0–10% for MiniMax-M3 queried statelessly. All agent-based readings reported in this paper therefore use a Latin-square design in which each agent sees each item in only one form (Sec. H).

## D. Negative Results

**Global detector.** A detector that also flagged four-digit numbers and long letter-only tokens (later replaced by the one in Sec. 3.2) flagged on average 85 and 112 words per MultiFieldQA and HotpotQA document, 62% of them years. On one LongBench sample it lowered F1 by 0.05 on average, with 3 of 12 cells significantly negative; on a second sample it was neutral (−0.005, 1 of 12 negative). With the final detector the difference to uniform rendering is +0.000 (0 of 12 negative) on the second sample.

**Word wrapping.** Wrapping prose at word boundaries wastes part of every line; on long documents at the 5 px budget this left no room to enlarge anything, so that variant matched or exceeded the uniform cost while gaining little (+0.00 to +0.12 recall). Character wrapping for prose (Sec. 3.3) removes this waste.

**Local instead of global shrinking.** Shrinking only the prose within $K$ words of a flagged string, with the smallest feasible $K$, was harmless on LongBench (+0.003, 0 of 12 negative) but retained only 84% of the RULER gain at the 6 px budget, because it fills the whole budget with slightly smaller enlarged strings. It brought no benefit over global shrinking.

**Other variants.** On short manifests, enlarging only the hash and path substrings of a manifest line freed almost no space. Adding one pixel of line spacing to prose-only lines did not change prose accuracy (+0.00, 95% CI $[-0.09, 0.09]$). Keeping prose at the uniform size left room only for 6–8 px identifiers and recovered 13–24 points less than full reallocation.

## E. Hazard Density and Routing

Figure 5 varies the number of flagged strings per 32K-character document from 1 to 256 (per-reader values, including Claude, in Tab. 8). Allocation alone degrades as density grows (0.42 at $k$=1, 0.24 at $k$=16, 0.06 at $k$=64), because more strings compete for the pixels prose can give up. Anchors are strong while hazards are sparse but become both less accurate and expensive as they multiply, reaching 0.34 recall (K) at $k$=64 for 1.8× the tokens of uniform rendering. At $k$=4, X recovers 0.98 of the needles while costing 13% less than uniform rendering and 20% less than K. Under the precision-aware F1, however, X and K do not differ (pooled over $k \in \{1, 4, 16\}$: −0.01, CI [−0.07, 0.04]), and several readers return most of the anchor list at $k \geq 4$. For routing, $\tau$ was chosen per reader by two-fold cross-validation on this sweep; the selected values have median 0.3. Sweeping $\tau$ traces the recall–cost frontiers of panel (c), on which routing with X matches or dominates routing with K; at $\tau$=0.3 it reaches 0.97 recall at 0.69× the plain-text cost versus 0.93 at 0.70×. These frontiers are computed

Table 7. Short documents (≈1.6K characters, one image, $N$=30). Difference between word-wrapped reallocation and uniform rendering in identifier exact-match (ids) and in accuracy on three prose probes (prose); column headers give data and characters per vision token. $^{*}$/$^{\dagger}$: 95% CI above/below zero.

| | synthetic, 21 | | synthetic, 14 | | manifests, 14 (a) | | manifests, 14 (b) | |
|---|---|---|---|---|---|---|---|---|
| Reader | ids | prose | ids | prose | ids | prose | ids | prose |
| Claude Sonnet 5.5 | +0.35$^{*}$ | +0.07 | +0.19$^{*}$ | +0.06 | +0.21$^{*}$ | -0.10 | +0.13$^{*}$ | -0.08 |
| Gemini 3.5 Flash-Lite | +0.33$^{*}$ | +0.01 | +0.04$^{*}$ | -0.01 | +0.23$^{*}$ | +0.00 | +0.19$^{*}$ | -0.01 |
| GPT-6 Luna | +0.21$^{*}$ | +0.06 | +0.41$^{*}$ | +0.03 | +0.17$^{*}$ | -0.16$^{\dagger}$ | +0.18$^{*}$ | +0.03 |
| Qwen3-VL-235B | +0.04$^{*}$ | +0.01 | +0.44$^{*}$ | -0.08 | +0.06$^{*}$ | -0.04 | +0.08$^{*}$ | -0.00 |
| Llama 4 Maverick | +0.05$^{*}$ | +0.00 | +0.66$^{*}$ | -0.39$^{\dagger}$ | +0.18$^{*}$ | -0.38$^{\dagger}$ | +0.27$^{*}$ | -0.38$^{\dagger}$ |
| MiniMax-M3 | +0.28$^{*}$ | -0.03 | +0.57$^{*}$ | -0.02 | +0.17$^{*}$ | -0.21$^{\dagger}$ | +0.21$^{*}$ | -0.14$^{\dagger}$ |

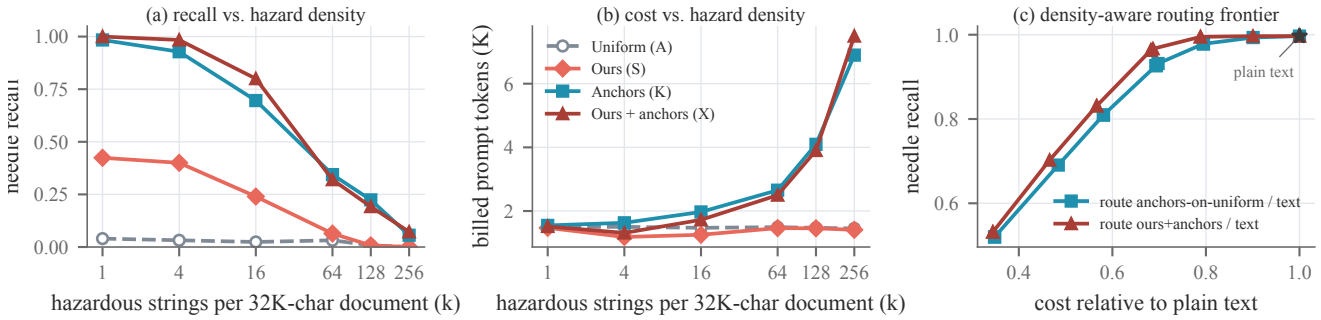



Figure 5. **Hazard density.** 32K-character documents with $k$ UUID needles, one of which is queried ($N$=25). (a) Recall and (b) billed tokens for uniform rendering (A), our allocation (S), anchors on uniform rendering (K), and our allocation with anchors (X); recall averages the five non-Claude readers; cost averages the readers that report usage (Claude included, MiniMax excluded). (c) Recall versus cost relative to plain text when each document is routed either to an image with anchors or to plain text by Eq. (3), sweeping $\tau$ (the four readers with billed costs for both images and plain text).

on the same sweep used to select $\tau$ and should be read as in-sample.

Table 8. Per-reader recall in the hazard-density sweep, K/X (anchors on uniform rendering / on our rendering).

| Reader | $k$=1 | 4 | 16 | 64 | 128 | 256 |
|---|---|---|---|---|---|---|
| Claude Sonnet 5.5 | 1.00/1.00 | 1.00/1.00 | 0.88/0.96 | 0.60/0.72 | 0.28/0.44 | 0.16/0.24 |
| Gemini 3.5 Flash-Lite | 0.96/1.00 | 0.96/1.00 | 0.68/0.88 | 0.36/0.28 | 0.28/0.16 | 0.08/0.08 |
| GPT-6 Luna | 0.96/1.00 | 0.88/1.00 | 0.60/0.92 | 0.32/0.32 | 0.20/0.12 | 0.12/0.12 |
| Qwen3-VL-235B | 1.00/1.00 | 1.00/1.00 | 1.00/1.00 | 0.04/0.04 | 0.00/0.00 | 0.00/0.00 |
| Llama 4 Maverick | 1.00/1.00 | 0.80/0.96 | 0.48/0.44 | 0.48/0.20 | 0.16/0.16 | 0.08/0.04 |
| MiniMax-M3 | 1.00/1.00 | 1.00/0.96 | 0.72/0.76 | 0.52/0.76 | 0.48/0.52 | 0.00/0.12 |

## F. Natural-Identifier Questions

For each of six LongBench tasks we shuffled the documents of at most 40K characters with a fixed seed and kept the first ten that contain a usable target. A target is a word flagged by the detector of Sec. 3.2 that, after stripping surrounding punctuation, has 6–64 ASCII characters including a digit, starts with a letter or digit, and is preceded by six words none of which is flagged; the six-word phrase must occur exactly once in the document. We exclude dataset artifacts: Qasper reference placeholders (e.g., `BIBREF`), formula fragments, characters replaced by "?" during ASCII normalization, and words glued across a missing space (a comma directly followed by a letter, a period or closing parenthesis followed by an uppercase letter, and a few letter–digit patterns); targets containing "=" are also excluded. A first version of the set without the last two filters contained such artifacts in 14 of its 60 questions; because anchors and enlarged rendering present the glued string as a single unit, that version favored our arms, and we discarded it. The prompt asks for the exact string that follows the phrase; scoring is the same recall as for RULER. The same 60 questions were also rendered at base sizes of 5 and 7 px for the frontier of Sec. 5.5.

**Detector-independent set.** To measure what happens to strings the detector misses, a second set (new seed) samples the target uniformly from all tokens of 4–64 ASCII characters that contain a digit, with the same phrase and artifact filters (including `EQREF` and `DISPLAYFORM` placeholders) but without requiring the detector to flag the target or the document. The detector flags one of the 60 targets.

## G. Real-Corpus Profile

Table 9 lists the routing profile of real corpora: 13 English LongBench tasks (20 chunks each), Paul Graham essays, Python source files from an installed package environment, local system logs, and Python package `RECORD` manifests (36–40 chunks each). Every chunk is rendered and reallocated so that $r$ uses the actual image cost.

Table 9. Routing profile of real corpora (32K-character chunks, $\tau$=0.3). Flags per 1K characters, median and 90th-percentile routing ratio $r$, share routed to image+anchors (X), and estimated cost relative to plain text. Text tokens are estimated with a linear fit to billed text-only reads (0.208 tokens per ordinary and 0.737 per flagged character; mean absolute relative error 1.1%).

| Corpus | $n$ | flags/1K | $r$ med | $r$ p90 | →X | cost/T |
|---|---|---|---|---|---|---|
| LB/narrativeqa | 20 | 0.06 | 0.00 | 0.01 | 95% | 0.27 |
| LB/qasper | 20 | 0.18 | 0.03 | 0.13 | 100% | 0.19 |
| LB/multifieldqa_en | 20 | 0.85 | 0.05 | 0.98 | 75% | 0.43 |
| LB/hotpotqa | 20 | 0.17 | 0.01 | 0.11 | 95% | 0.25 |
| LB/2wikimqa | 20 | 0.06 | 0.01 | 0.05 | 100% | 0.21 |
| LB/musique | 20 | 0.06 | 0.00 | 0.02 | 100% | 0.22 |
| LB/gov_report | 20 | 0.04 | 0.00 | 0.02 | 100% | 0.21 |
| LB/qmsum | 20 | 0.00 | 0.00 | 0.00 | 100% | 0.23 |
| LB/multi_news | 20 | 0.06 | 0.00 | 0.05 | 100% | 0.21 |
| LB/triviaqa | 20 | 0.23 | 0.08 | 0.19 | 95% | 0.23 |
| LB/passage_retrieval_en | 20 | 0.04 | 0.00 | 0.02 | 100% | 0.21 |
| LB/lcc | 20 | 1.40 | 0.40 | 0.83 | 25% | 0.81 |
| LB/repobench-p | 20 | 1.43 | 0.52 | 0.87 | 5% | 0.96 |
| pg_essays | 36 | 0.01 | 0.00 | 0.00 | 100% | 0.22 |
| python_source | 40 | 2.62 | 0.67 | 1.47 | 12% | 0.91 |
| system_logs | 40 | 11.84 | 3.73 | 4.09 | 0% | 1.00 |
| pkg_manifests | 40 | 11.32 | 16.09 | 16.72 | 0% | 1.00 |

## H. Additional Claude Reader

We also used Claude through an agent interface as a reader that is never used to choose any setting. For $k \in \{1, 4, 16, 64, 256\}$, ten items of the density sweep were read as X, K, and plain text in a Latin square: 15 agents each read ten different items, each in one form, using only a file-reading tool and a tool to write their answer file. Costs are estimated with Eq. (1) for images and with the linear text-token fit of Tab. 9. Table 10 shows that this stronger reader tolerates higher density than the API readers (0.90 for X at $k$=64, against 0.60 for K), and with $\tau$=0.3 the routed system reaches 0.98 recall at an estimated 0.61× the plain-text cost. Agents may resample images when reading them, and $N$=10 per cell is small, so these numbers are indicative.

Table 10. Additional Claude reader (agent interface, one item per form per agent; $N$=10 per cell). Recall and estimated cost relative to plain text.

| $k$ | X | K | T | cost X/T | cost K/T |
|---|---|---|---|---|---|
| 1 | 1.00 | 1.00 | 1.00 | 0.23 | 0.23 |
| 4 | 1.00 | 1.00 | 1.00 | 0.17 | 0.24 |
| 16 | 0.90 | 1.00 | 1.00 | 0.22 | 0.27 |
| 64 | 0.90 | 0.60 | 1.00 | 0.36 | 0.36 |
| 256 | 0.20 | 0.00 | 1.00 | 0.59 | 0.60 |

## I. Reproducibility and Ethics

**Prior work.** The detector design, anchors, and text routing follow [1], which is under review; an anonymized copy is included in the supplementary material.

**Code and data.** The renderer, detector, router, task generators, reader clients, analysis scripts, all logged model outputs, and the scripts that produce every figure and table will be released. Task data are generated from public sources (RULER templates with Paul Graham essays, LongBench) with fixed seeds.

**Compute and cost.** No model was trained. All readers were accessed through provider APIs between November 1 and 4, 2026; reader versions are identified by their provider model names in Sec. 4. Pay-per-token API charges for all experiments, including pilots, were about $55; MiniMax-M3 and the agent-based Claude reader were used under subscription plans.

**Ethics.** The study involves no human subjects and no personal data. The system logs profiled in Sec. G were processed locally to count flagged strings; their contents are not released. Better verbatim fidelity under compression could in principle help extract identifiers from rendered documents, but the method only changes how a document the user already holds is presented to a model.